Lab 7: Fuzzing for Input Validation Bugs

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Objective

Use Hypothesis to fuzz the three functions in phone_repair_quote_parser.py (mask_imei, parse_parts_order and parse_model), find inputs that crash or misbehave, fix them, and confirm the fixes with tests.

Bugs found and fixes

- **mask_imei:** crashed on None, and showed values of four characters or fewer unmasked. Fix: return an empty string for non-string input and fully mask short values.
- **parse_parts_order:** crashed on items without an "x" (such as "screen"), non-numeric quantities, part names containing "x" and None. Fix: split on the last "x", skip malformed items and return an empty dictionary for non-strings.
- **parse_model:** crashed on a single word, tab-separated input, an empty string and None. Fix: split on any whitespace, return (brand, "") for one word and ("", "") for empty input.

Screenshots

Screenshot 1: Original module. "12" is not masked, and "screen" crashes parse_parts_order with a ValueError.

```
NIDHI S@laptop-Tachibana MINGW64 ~/PES1UG24AM919_SE-LAB/Lab7_Phone_Repair_Quote (main)
$ python phone_repair_quote_parser.py
==== Running mask_imei ====
Enter an IMEI number (e.g. 35-209900-176148-1): 12
Masked IMEI: 12

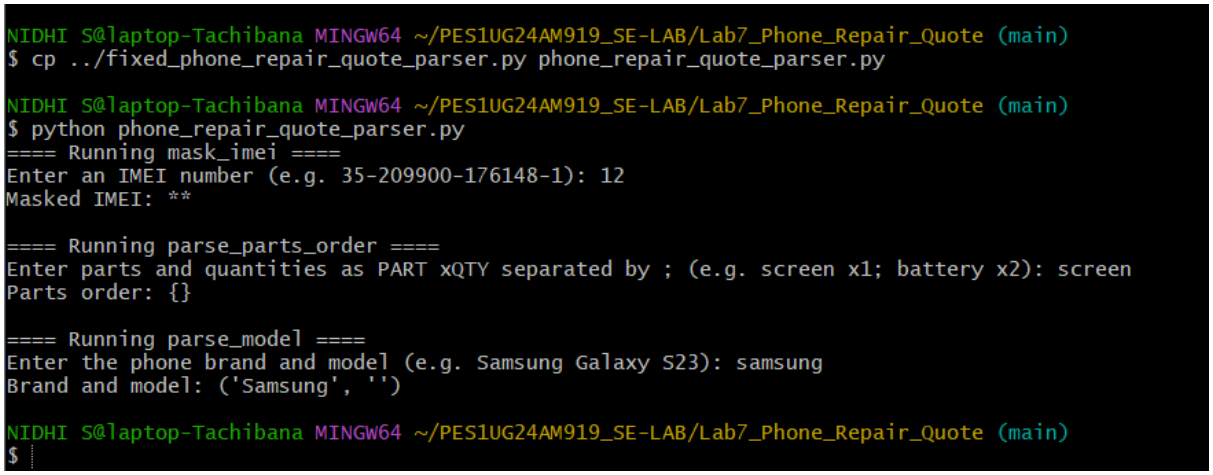
==== Running parse_parts_order ====
Enter parts and quantities as PART xQTY separated by ; (e.g. screen x1; battery x2): screen
Traceback (most recent call last):
  File "C:\Users\NIDHI S\PES1UG24AM919_SE-LAB\Lab7_Phone_Repair_Quote\phone_repair_quote_parser.py",
 line 42, in <module>
    main()
  File "C:\Users\NIDHI S\PES1UG24AM919_SE-LAB\Lab7_Phone_Repair_Quote\phone_repair_quote_parser.py",
 line 35, in main
    print("Parts order:", parse_parts_order(value))
                          ^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\NIDHI S\PES1UG24AM919_SE-LAB\Lab7_Phone_Repair_Quote\phone_repair_quote_parser.py",
 line 16, in parse_parts_order
    part, qty = item.split('x')
    ^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
```

Screenshot 2: Hypothesis tests against the original module: 3 failed.

```
    +------------------------------------
=========================== short test summary info ============================
FAILED test_phone_repair_quote_parser.py::test_mask_imei_no_crash - AssertionError: mask_imei crashe
d on None: AttributeError: 'NoneType' object has no attribute '...
FAILED test_phone_repair_quote_parser.py::test_parse_parts_order_no_crash - ExceptionGroup: Hypothes
is found 3 distinct failures. (3 sub-exceptions)
FAILED test_phone_repair_quote_parser.py::test_parse_model_no_crash - ExceptionGroup: Hypothesis fou
nd 2 distinct failures. (2 sub-exceptions)
============================== 3 failed in 6.95s ===============================

NIDHI S@laptop-Tachibana MINGW64 ~/PES1UG24AM919_SE-LAB/Lab7_Phone_Repair_Quote (main)
$
```

Screenshot 3: Fixed module with the same inputs. No crash, and "12" is now masked.



Screenshot 4: Hypothesis tests against the fixed module: 3 passed.

